Titus Loftin

McKetta Department of Chemical Engineering
University of Texas at Austin
loftintitus@utexas.edu

Education

The University of Texas at Austin *Expected Graduation: 2027*
Bachelor of Science in Chemical Engineering GPA: 3.84 / 4.00

Relevant Coursework

- Transport Phenomena (Momentum, Heat, and Mass Transfer)
- Chemical Engineering Thermodynamics
- Numerical Methods for Engineers
- Probability and Statistics for Engineering
- Genetics
- Organic Chemistry I & II

Research Experience

Soft Tissue Biomechanics Lab, University of Texas at Austin Austin, TX
Undergraduate Researcher September 2024 – Present

- Working under Dr. Manuel Rausch.
- Developed the Experimental Mechanical MNIST dataset to enable machine learning approaches for heterogeneous materials, supporting analysis of biologically complex tissues.
- Designed an in vitro **flow-loop system** to study hemodynamic-driven clot embolization.
- Modeled embolization as a **fatigue-like process**, defining a probability distribution for embolization in the popliteal vein.
- Quantifying the effects of **clot composition** on embolization threshold and failure mode.
  - *Two First-author manuscripts in preparation (ML mechanics; embolism risk)*

Publications

- Tac, V., Amiri, A., Bechtel, G.N., **Loftin, T. ...**, Tepole, A. Fully data-driven inverse characterization of heterogeneous materials with hyper-network neural ODEs. *npj Computational Materials*, in review.

Presentations

- **Loftin, T.**, et al. "A Benchmark Dataset for Machine Learning and Data-Driven (Inverse) Approaches in Mechanics." *USNCCM18*, July 2025.
- **Loftin, T.**, et al. "Machine Learning and Data-Driven Approaches in Mechanics." UT Austin Week of Research, March 2025.

Leadership and Service

Chemical Engineering Undergraduate Advisory Board
Founder and Director Fall 2024 – Present

- Founded and led the undergraduate advisory board to advocate for student access, inclusion, and equity in departmental decision-making.
- Raised membership to over 120 students, representing approximately 20% of the undergraduate population.

Chemical Engineering Faculty Selection Committee
Student Member Spring 2025, Spring 2026

- Serve as an undergraduate representative in faculty hiring discussions and candidate evaluations.

Tau Beta Pi, Texas Alpha Chapter
Vice President of Corporate Relations Spring 2025 – Present

- Established 10 corporate partnerships totaling over $7,000 to support chapter activities.
- Manage and lead several officers and approximately 35 candidates per semester.

Additional Community Service
Volunteer 2024 – Present

- Volunteered with Baylor Scott & White, Ascension Seton, Healing with Horses Ranch, and the Central Texas Down Syndrome Association, supporting patient care, accessibility, and community programs.

Teaching Experience

University of Texas at Austin
Learning Assistant, CHE 354 & CHE 319 January 2026 – Present

- Learning Assistant for undergraduate chemical engineering core courses covering unit operations and transport phenomena.
- Supported students through office hours, recitations, problem set development, and grading.

University of Texas at Austin
Undergraduate Tutor, CHE 210 August 2025 – Present

- Provided individual and group tutoring in material balances and spreadsheet-based analysis.

Selected Personal Projects

Autonomous Vital Recording Robot

- Designed and built an autonomous system to measure blood pressure, pulse, and blood oxygen saturation.
- Selected as one of 600 applicants nationwide to receive project funding through the MIT THINK Scholars Program.

No-Cost MCAT Studying Web Application

- Designed and deployed a web-based study platform providing free access to MCAT flashcards, practice questions, and content review materials.
- Built the application to reduce financial barriers to standardized test preparation through accessible, no-cost educational tools.

Awards and Honors

- Undergraduate Research Experiences Grant, Fall 2024 – Present
- University Honors, Fall 2024 – Present
- New Engineering Solutions for Tomorrow Competition, 1st Place, 2025
- Tod Joe Chemical Engineering Endowed Scholarship, 2025 – 2026
- Dell Scholar, 2024 – 2027
- MIT THINK Scholar, 2024
- Texas Exes Austin Area Scholar, 2024

Skills

- **Computational & Data Science:** Python (PyTorch, NumPy, SciPy, scikit-learn), MATLAB; convolutional neural networks (CNNs), Fourier neural operators (FNOs), inverse modeling, probabilistic modeling, data-driven mechanics
- **Modeling & Analysis:** Continuum mechanics, nonlinear material modeling, force–displacement analysis, full-field strain mapping, statistical analysis of experimental data
- **Experimental & Wet Lab:** Mechanical testing of soft biological tissues, in vitro flow-loop experiments, clot fabrication and characterization, chromatography, electrophoresis, PCR
- **Design & Fabrication:** SolidWorks, Fusion 360; experimental system design, rapid prototyping, sensor integration